

Ditsa Chakraborty

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PROFILE SUMMARY

Electronics & Communication Engineering student with hands-on experience across AI/ML, data science, embedded systems, and software development. Took ownership of building and deploying production fintech applications during an industry internship, handling everything from data analysis to CI/CD deployment. Also worked on ESP32 based embedded communication systems and validated ML models (Neural Networks, Random Forest, CNN based computer vision) at 90%+ accuracy.

EDUCATION

B.Tech., Electronics & Communication Engineering SRM Institute of Science and Technology CGPA: 8.34 / 10 (up to 6th Semester)	2023 – 2027 Chennai, TN
Std XII — CBSE Indus Valley World School 75.2%	2022 Kolkata, WB
Std X — CBSE Indus Valley World School 86%	2020 Kolkata, WB

INDUSTRY EXPERIENCE

InvestTech Analyst Intern ZeTheta Algorithms Pvt. Ltd.	02/2026 – 05/2026 Remote
- Owned end-to-end delivery of four production fintech applications (Market Depth Explorer, Convertible Bond Pricing Framework, Market Monitoring Software, ML-Based Sentiment Analysis Trading Platform) from requirements through deployment, coordinating across design, engineering, and QA. - Drove process improvements in the model validation pipeline, reducing turnaround time on statistical model checks and improving reliability of trading signal outputs. - Built and maintained CI/CD deployment workflows (Render, Vercel) and Git-based version control processes adopted across the intern team, improving release consistency and reducing deployment errors. - Performed quantitative analysis and statistical modeling using Python and Pandas; built an NLP pipeline using FinBERT to analyze sentiment, generate trading signals, and run simulations. - Delivered full ownership transfer of all four repositories to the company, including a recorded walkthrough and demo of each project (linked via unlisted YouTube video in the README) to support handoff.	

VOLUNTEER EXPERIENCE

Volunteer, My Children Foundation (NGO)	06/2025 – 07/2025 Kolkata, WB
Contributed to digital outreach initiatives supporting underprivileged children, including editing and transcribing videos for social media and fundraising campaigns.	

ACADEMIC PROJECTS

Transmission and Reception of Signal Using ESP-32 with Machine Learning SRM Institute (Team)	05/2026 Chennai, TN
- Built and deployed a multi-nodal communication system using ESP32 devices for signal transmission and reception experiments. - Conducted performance experiments to gather and analyze RSSI data under various conditions and beams. - Used Neural Network and Random Forest classification algorithms to predict communication behavior with 99.7% and 90.65% accuracy, respectively. - Built a real-time Python dashboard for tracking communication data. - Presented work at RASET 2026 – a National Conference organized by IEEE OES.	
Fruit Quality Assessment using MobileNetV2 SRM Institute	04/2026 Chennai, TN
- Designed and implemented a Convolutional Neural Network (CNN) machine vision pipeline using MobileNetV2 with transfer learning for classifying fresh vs. spoiled fruit, achieving 96.1% accuracy. - Conducted data pre-processing, model training, testing, and validation; completed technical documentation and material for an IEEE publication.	

TECHNICAL SKILLS

Data Analytics & Machine Learning: Pandas, Scikit-Learn, Data Analysis, Data Visualization, Plotly, Neural Networks, Random Forest, MobileNetV2, FinBERT, NLP
Embedded Systems & ECE: ESP32, Arduino, FPGA/Verilog, IoT, LTspice, Scilab
Programming Languages: Python, SQL, Java, TypeScript, JavaScript
Databases & Cloud: PostgreSQL, Supabase, Vercel, Render

Software Development: React, Node.js, Express.js, REST APIs, Git, CI/CD
Operations & Process: Cross-functional Coordination, Process Optimization, Project Execution
Professional Skills: Research, Technical Documentation, Problem Solving, Presentation Skills, Team Collaboration, Design Thinking

CERTIFICATIONS & COURSES

Goldman Sachs Operations Job Simulation — Forage	08/2026
Certificate of Presentation — RASET 2026 (IEEE OES) <i>Bannari Amman Institute of Technology</i>	05/2026
Python for Data Science, NPTEL <i>IIT Madras</i>	07/2024 – 08/2024
AI Foundations for Everyone, Coursera <i>IBM</i>	04/2025
Design Thinking – A Primer, NPTEL <i>IIT Madras</i>	01/2025 – 02/2025
Certificates of Project Work <i>Phantom Liquidity Hunter, Convertible Bond Laboratory, Market Depth Explorer, Sentiment Analysis Trading — ZeTheta Algorithms</i>	03/2026 – 05/2026

ACHIEVEMENTS

First Prize — Cooking Without Fire, 15th Annual Cultural Competition	2024
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